Mohammad Shaheer

mshaheer@alumni.cmu.edu, mshaheer@andrew.cmu.edu | +1(412)287-1845, +97450507802

Education

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Computer Science - Research Thesis — May 2027

Carnegie Mellon University — **Doha, Qatar**
Bachelor of Science in Computer Science | Concentration in Algorithms & Complexity — May 2023
University Honors & College Honors
Relevant Coursework: Algorithms, Constructive Logic, Computer Security, Machine Learning, Computer Systems, Distributed Systems, Robot Learning, Planning and Decision making in Robotics

Skills

Programming Languages: Python, C/C++/C#, Java, SML, Typescript/Javascript
Frameworks & Databases: Django REST Framework/Django, .NET, Flutter, NextJS, MongoDB, PostgreSQL, React
Cloud & Tools: GCP, AWS, Azure, Heroku, Docker, Kubernetes, Isaac Lab/Isaac Sim, ROS2, Gazebo, rviz, Linux

Professional Experience

AirLab - Carnegie Mellon University | Robotics Researcher | Pittsburgh, PA — May. 2026 - Present
- Built Isaac Lab RL environments across multiple task domains, **enabling training and evaluation of humanoid skills**
- Used coding agents to **generate and refine perceptual predicates** for long-horizon task and motion planning
- Trained low-level skills in simulation **to provide reusable behaviors** for accomplishing higher-level tasks
- Collected motion-capture and RGB data to build a real-to-sim pipeline **for training humanoid skills**

Avey | Full Stack Software Engineer | Doha, Qatar — Dec. 2023 - May. 2026
- Developed Coder, an AI-powered insurance claim auditing platform to **automate error detection and reduce claim rejections**; deployed in 2 local hospitals supporting 4 insurance companies
- Engineered a token escrow system to prevent overcharging; **reducing monthly billing errors from ~30% to ~3%**
- Replaced ad hoc Slack-based approval processes with a centralized, audited system, **cutting average approval time from days to hours across ~100+ monthly requests**
- Optimized Django ORM queries on a high-traffic list endpoint, **reducing response time from ~900ms to ~45ms** by profiling query execution and eliminating N+1 patterns

Snoonu | Software Engineer | Doha, Qatar — May 2023 - Dec 2023
- Conducted EDA on marketing campaign data to assess feature suitability for demand prediction models
- **Improved testing accuracy of demand forecasting model by ~17%** through hyperparameter tuning
- Deployed a real-time driver tracking system communicating driver status updates to the merchant portal periodically via SignalR and RabbitMQ **reducing merchant calls to operations team by 37%**

Projects

Formal Verification of Smart Contracts — Carnegie Mellon University | Jul. 2022 - Jan. 2026
- Designed **oracle and stablecoin blockchain protocols** satisfying properties related to **economic fairness, long-term sustainability and stability**
- Implemented the protocols in **Solidity** and formally verified desirable properties using the **Rocq** theorem prover
- Oracle paper accepted for the **TYPES 2023** conference; Stablecoin paper accepted for the **Stability 2026** workshop

Distributed File System | Java — Carnegie Mellon University | Nov. 2022
- Built a distributed file system using custom RMI with multi-threaded Skeleton/Stub architecture, **enabling remote file operations across multiple storage servers**
- Implemented a metadata management system on a centralized naming server, coordinating storage server registration, duplicate pruning, and synchronized create/delete operations across nodes
- Implemented hierarchical read-write locking with FIFO-ordered queues and root-to-leaf path locking, **ensuring consistency for concurrent multi-client reads and writes while preventing deadlocks**
- Engineered a dynamic replication engine that auto-scaled file replicas based on request load using a linear coarse-grained policy, with invalidation-based write propagation **to keep replicas consistent**